EduPulse AI

Student Success Early-Warning and Intervention Platform

AI Mini Group Project · Academic prototype localized for a Kabarak University context

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COMP 413 · Dr Andrew Kipkebut · 17 August 2026

The Real-World Problem

Student difficulty may be recognized too late

Semester 1 signals can support earlier review

Purpose: academic, financial and wellbeing support—not penalties

Objective and AI Task

Supervised multiclass classification

Outcomes: Dropout · Enrolled · Graduate

Prediction point: end of Semester 1

Outputs: estimated probabilities, support priority, signals and pathways

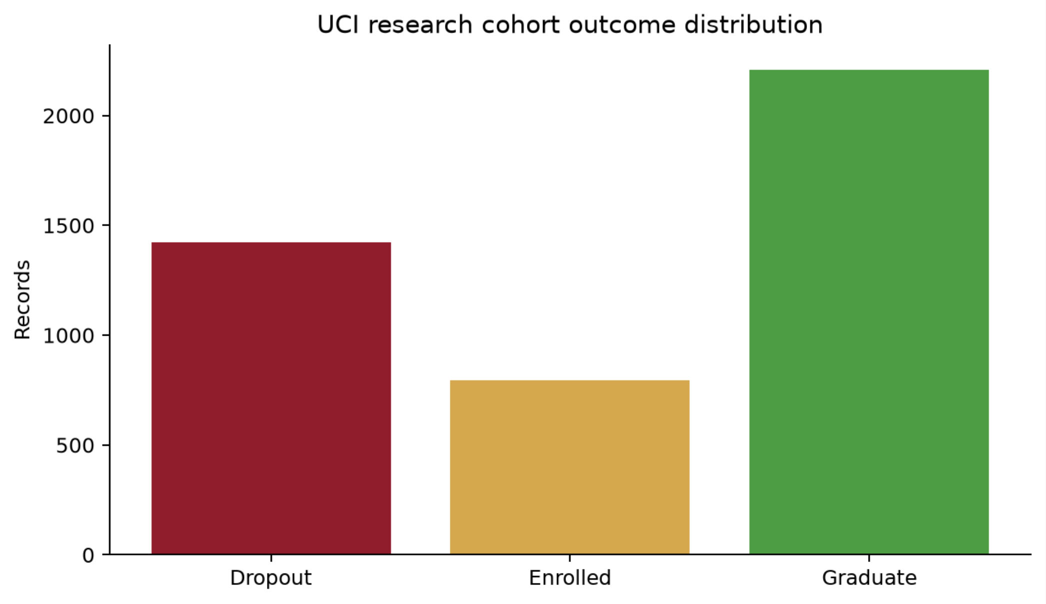
Dataset: Transparent Research Evidence

UCI Dataset 697 · 4,424 records · 36 predictors

Dropout 1,421 · Enrolled 794 · Graduate 2,209

Portuguese higher-education context · CC BY 4.0

Not Kabarak University student data



Preparation and Leakage Control

0 missing cells · 0 duplicates · expected target classes

Stratified 80/20 split · seed 42

All selection and tuning on training folds only

Semester 2, identity and sensitive/proxy fields structurally excluded

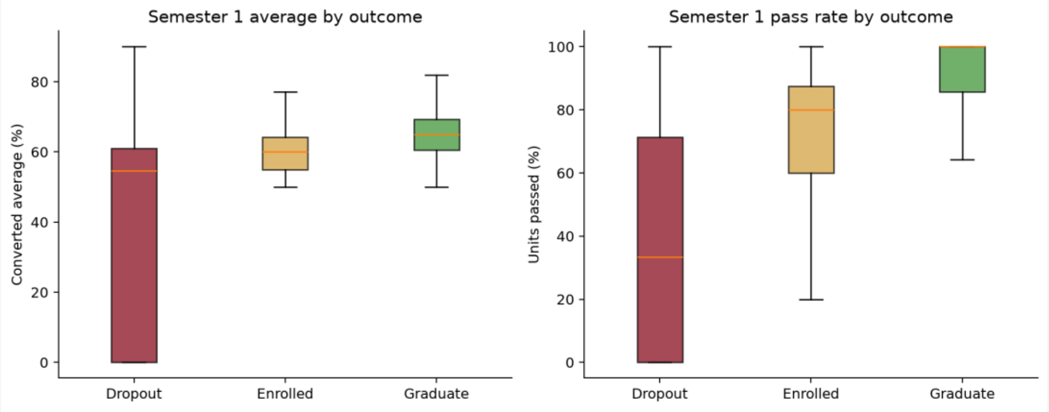
Exploratory Data Analysis

Semester 1 marks and pass rates shift across outcomes

Distributions overlap: no single input is decisive

Smaller Enrolled class makes accuracy insufficient

Associations do not establish causes



Compact Feature Engineering

Seven lecturer-friendly inputs

Derived: pass rate, units not passed, assessments per unit

Guarded division and range validation

0–20 UCI grade × 5 for a 0–100 display

Engineered compact CV Macro F1: 0.662

Model Development

Logistic Regression · Random Forest · Extra Trees

Support Vector Machine · Histogram Gradient Boosting

5-fold stratified CV · primary metric Macro F1

Top two models tuned with bounded randomized search

Final Results

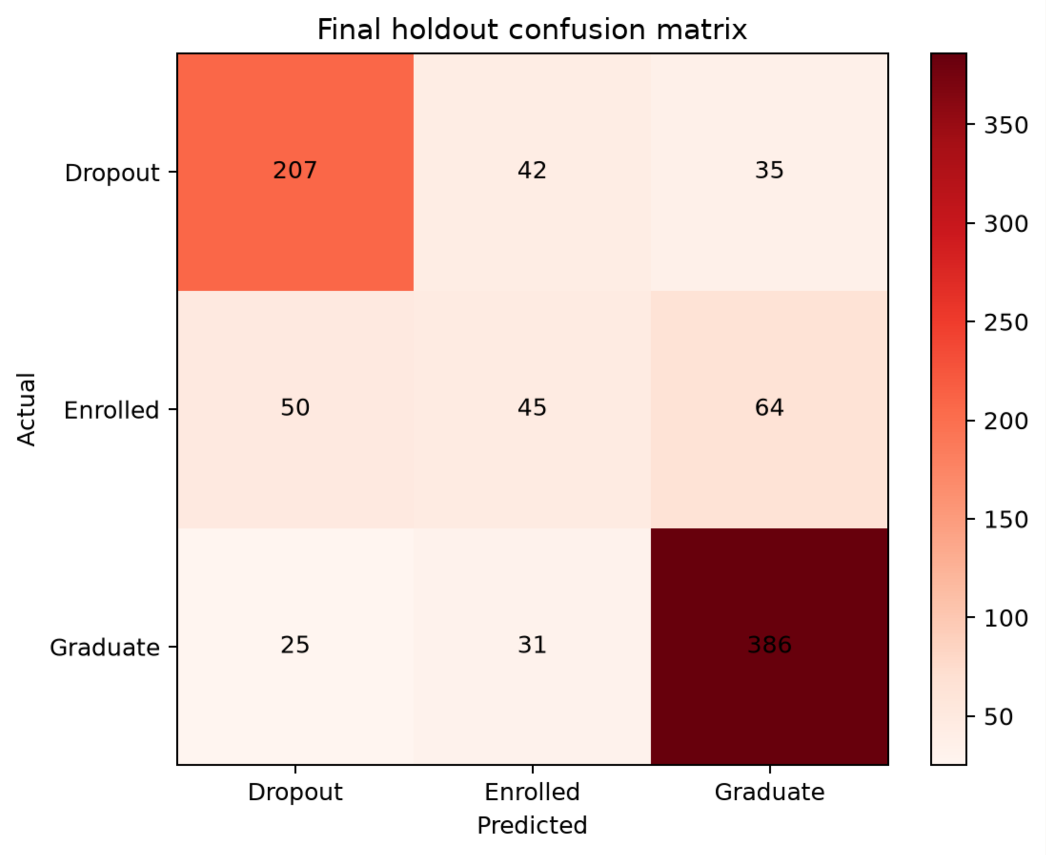
Selected: Support Vector Machine

CV Macro F1: 0.665

Holdout Macro F1: 0.630

Balanced Accuracy: 0.628 · Accuracy: 0.721

Dropout recall: 72.9%



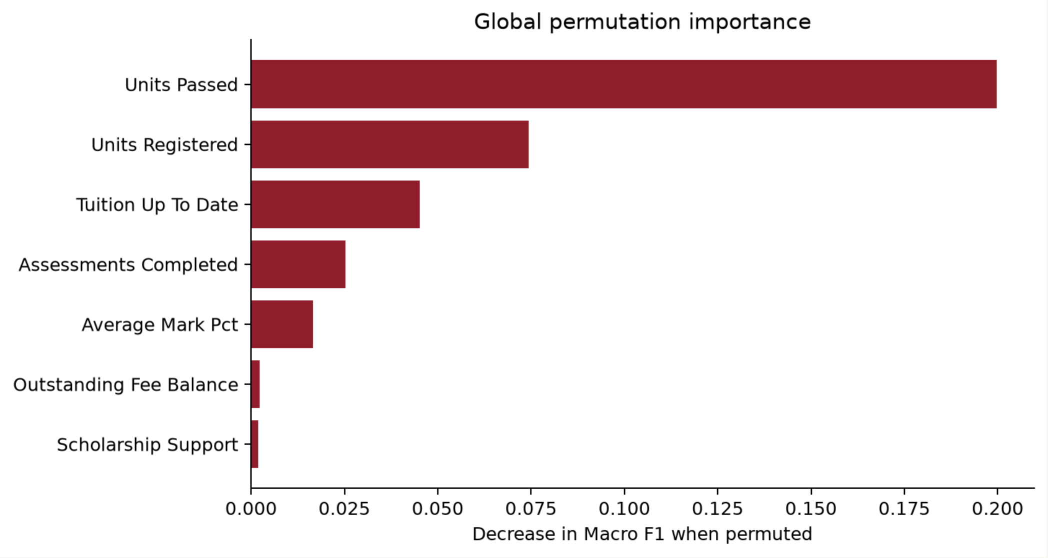
Explainability and Key Insight

Global permutation importance measures model reliance

Individual sensitivity compares with research medians

Language is associative, never causal

Probabilities are calibrated estimates, not guarantees



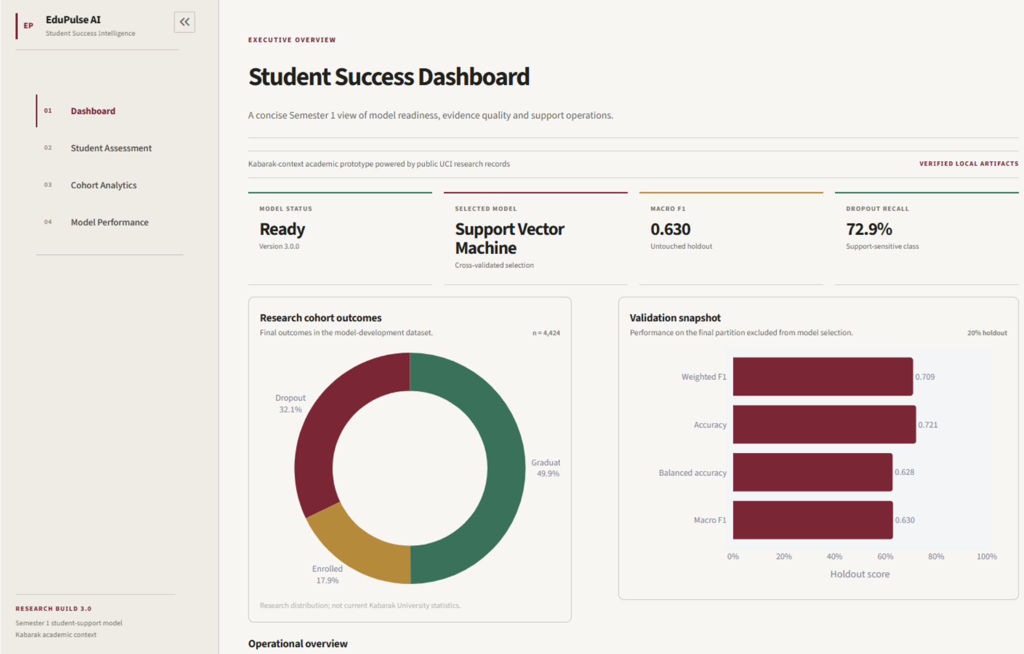
EduPulse AI Portal

Dashboard · Student Assessment

Cohort Analytics + validated batch assessment

Model Performance with real evidence

Local saved pipeline · one canonical results source



Live Demonstration

1. Load Higher Support Need demo profile
2. Run the seven-input assessment
3. Review priority, all probabilities and model signals
4. Show supportive pathway and Model Performance

Limitations, Future Work and Conclusion

No Kabarak validation; source context is Portuguese

Enrolled is the hardest outcome class

Next: governance, privacy, local validation, fairness and monitoring

A reproducible support tool—not an automated authority